MAT 181- Chapter 10 Practice Problems-Ans

Name_____Section:_____ Date:_____

Suppose you will perform a test to determine whether there is sufficient evidence to support a claim of a linear correlation between two variables. Find the critical values of r given the number of pairs of data n and the significance level α.

1) n = 14, α = 0.05

A) r = 0.553 B) r = ± 0.661 C) r = 0.532 D) r = ± 0.532

2) n = 45, α = 0.01

A) r = ± 0.294 B) r = ± 0.378 C) r = 0.402 D) r = 0.378

Given the linear correlation coefficient r and the sample size n, determine the critical values of r and use your finding to state whether or not the given r represents a significant linear correlation. Use a significance level of 0.05.

3) r = 0.523, n = 25

A) Critical values: r = ±0.487, significant linear correlation
B) Critical values: r = ±0.487, no significant linear correlation
C) Critical values: r = ±0.396, significant linear correlation
D) Critical values: r = ±0.396, no significant linear correlation

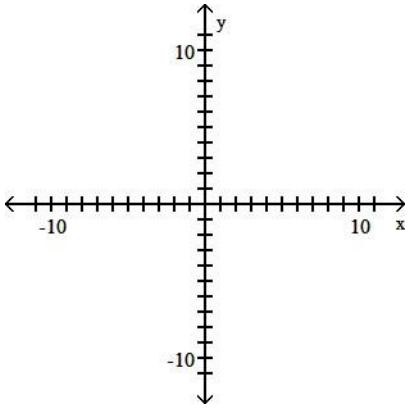
4) r = 0.843, n = 5

A) Critical values: r = ±0.878, significant linear correlation
B) Critical values: r = 0.950, significant linear correlation
C) Critical values: r = ±0.878, no significant linear correlation
D) Critical values: r = ±0.950, no significant linear correlation

Construct a scatterplot for the given data.

5)

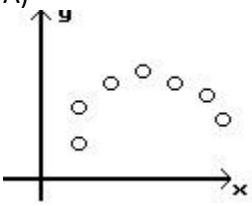
x	1	−7	−2	−4	1	5	−6	7	−4	−2
y	7	−1	−7	5	2	3	−2	1	−5	−3



Determine which scatterplot shows the strongest linear correlation.

6) Which shows the strongest linear correlation?

A)



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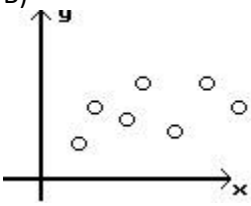

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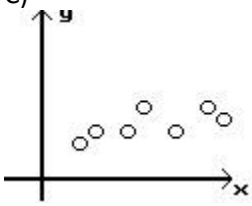


Find the value of the linear correlation coefficient r.

7) The paired data below consist of the test scores of 6 randomly selected students and the number of hours they studied for the test.

Hours	5	10	4	6	10	9
Score	64	86	69	86	59	87

Use the given data to find the best predicted value of the response variable.

8) Based on the data from six students, the regression equation relating number of hours of preparation (x) and test score (y) is $\hat{y} = 67.3 + 1.07x$. The same data yield $r = 0.224$ and $\bar{y} = 75.2$. What is the best predicted test score for a student who spent 2 hours preparing for the test?

9) The regression equation relating attitude rating (x) and job performance rating (y) for the employees of a company is $\hat{y} = 11.7 + 1.02x$. Ten pairs of data were used to obtain the equation. The same data yield $r = 0.863$ and $\bar{y} = 80.1$. What is the best predicted job performance rating for a person whose attitude rating is 73?

Use the given data to find the equation of the regression line. Round the final values to three significant digits, if necessary.

10)

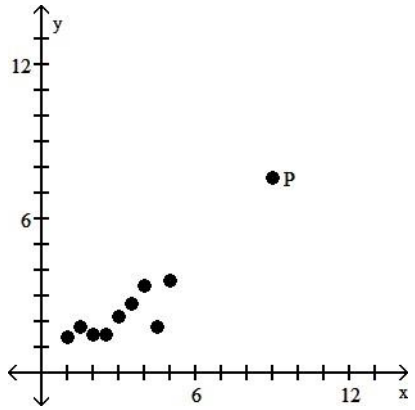
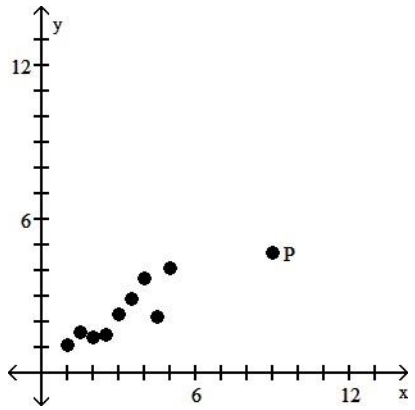
x	2	4	5	6
y	7	11	13	20

11)

x	6	8	20	28	36
y	2	4	13	20	30

Is the data point, P, an outlier, an influential point, both, or neither?

12)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